

A Retroactive Contact Channel for Behavior Cloning on Low-Cost Arms

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Abstract: Low-cost servo arms ship without force sensing, and standard imitation-learning stacks observe joint positions only, so learned policies close the gripper open-loop against contact. For logs that record synchronized joint-position commands and measured positions — which is what the public SO-100/SO-101 corpus already contains — the servo tracking error $\delta = a[t-1] - s[t]$ is a contact proxy computable after the fact, from fields those logs already carry. Current stacks discard it. In a pre-registered six-design study at 280 demonstrations on a simulated force-critical task (5 seeds $\times$ 100 paired episodes), every observation carrying this channel outperforms the channel-free base, while predicting δ as an auxiliary training target does not: the channel must be observed, not merely learned. The strongest design appends a free-motion-compensated residual, the *lag excess*, reaching 26.6% success against 6.2% (+20.4 points, $t = 4.7$), against a pre-declared resolution of 11.6 points at five seeds. A position-history control with no access to the channel reaches a comparable level, so this grid establishes that contact-relevant temporal information helps and not that this channel is uniquely capable. Because the lag excess reads one more frame of action history than a one-step baseline, we also ran an information-matched control; the advantage persists at +11.9 points (three seeds; 95% CI [+3.3, +20.7]), below the bar at that seed count. On 546 episodes across 16 public teleoperation datasets the pre-registered prediction that successful grasps carry higher δ fails: high δ at grasp time accompanies failure, consistent with operators forcing jammed grasps, which rules out mining such corpora for success labels through this channel alone. Code and data at [URL].

Keywords: Imitation Learning, Force Estimation, Low-Cost Manipulation

1 Introduction

Low-cost servo arms and open imitation-learning stacks have made manipulation research broadly accessible: ACT on ALOHA-class hardware [1], diffusion policies [2], and the LeRobot ecosystem [3] with its VLA descendants [4, 5, 6, 7] all train behavior cloning on teleoperated demonstrations, and all share one proprioceptive default: the current measured joint positions, one timestep, nothing else (Sec. 2).

This default fails at contact. The arms carry no force or tactile sensors, so a position-only policy cannot distinguish a seated grasp from an empty close; the failures are crushed objects, unseated grips, and drops during transport. The information needed to distinguish these cases is present in the logs. The servos are proportional position controllers, so when the jaw meets an object the measured position stalls while the command deepens, and the gap reflects the servo’s deflection under load. The authors of ACT state the physics directly: force “is implicitly defined by the difference between” leader and follower positions [1], and a recent intervention study shows on real hardware, across four force-critical tasks, that removing this discrepancy signal from ACT’s action labels breaks them [8].

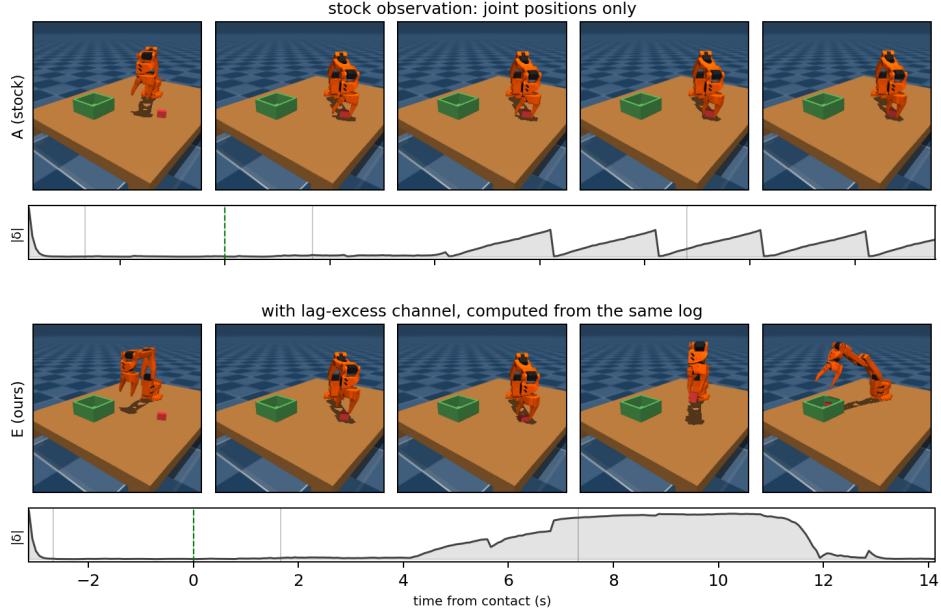


Figure 1: Two policies, identical scene and data budget (280 demonstrations). Top: the stock observation (joint positions only) reaches, grips, and releases without lifting. Bottom: the same network with the contact channel appended to its state vector places the block. The traces plot $|\delta|$ on the jaw joint; the contact plateau in both is the signal the top policy never receives. Both are computed from the same log fields.

39 Yet the stack keeps one operand of that subtraction in the action labels and never provides the
40 difference as an observation; the LeRobot Feetech driver reads only `Present.Position` into the
41 state vector, and the reference ACT implementation raises on `n_obs_steps != 1` [3].

42 The channel has a property no current- or torque-based estimator shares: it is recoverable retroac-
43 tively. Wherever a recorded dataset’s action field holds joint-position commands synchronized
44 with `observation.state` in a common frame, $\delta = a[t-1] - s[t]$ can be computed after the fact;
45 16 of the 20 public datasets we targeted passed our frozen inclusion gates (Sec. 4.3, which also
46 details the caveats on action semantics).

47 That property is what separates this channel from the alternatives, and the separation is concrete
48 rather than rhetorical. A recent survey of force sensing for manipulation learning organises the field
49 into families — finger-mounted optical, finger force, finger audio, joint torque, wrist force/torque,
50 and combinations — every one of which adds hardware. That survey states the premise of this work
51 in its own terms: “current robot foundation models focus exclusively on visual input and position
52 control. Relying on position alone might not be sufficient to achieve true dexterity, as small errors in
53 position may lead to large errors in force in stiff systems” [9]. It instances Gemini Robotics, 2000–
54 5000 episodes per task over twelve months, as not including force. Recovering the quantity from
55 logs never instrumented for it is not among the options it catalogues, and the phrase “tracking error”
56 does not appear in it. Of 123 archived SO-100/SO-101 datasets we could read, every one records
57 `observation.state` as joint positions alone — no load, no current. The remedy of Wong et al.
58 [8] needs motor current or joint effort, which the recording stack caps at half scale on the gripper
59 of every follower it connects (App. D); FACTR 2’s estimator infers external torque as the residual
60 between motor torque read from current and a predicted free-space torque, requiring no dedicated
61 force-torque sensor at collection or deployment [10] — but requiring current, which those logs do
62 not carry; Hwang et al. [11] needs a per-servo system identification performed on the hardware
63 itself. None of those requirements is satisfiable from a position-only log. The tracking error needs

64 two fields such logs already store. The same signal supports runtime enforcement from bus telemetry
65 alone (Sec. 4.4).

66 We ask what a small-data policy must observe to use this channel, and compare six observation
67 designs that differ only in which function of the logged pair enters the state vector. The best tested
68 form is the *lag excess* $e[t] = \delta[t] - \hat{K}r[t]$, which subtracts a per-joint free-motion following-lag
69 model fit on the jaw-open frames of the training demonstrations, label-free and frozen before training
70 (Sec. 3.1; implementation details in App. A).

71 Three results are established at our pre-declared seed resolution (Sec. 4.2). Channel-bearing obser-
72 vation beats channel-free in every input form. The tested auxiliary-prediction baseline recovers none
73 of the direct-input gain: the same information moves performance as an input and not as a training
74 target. And the lag excess is the only tested design to beat raw action history (26.6% vs. 13.8%,
75 CI [+2.8, +22.6]), though that baseline reads one fewer frame of action history than the lag excess
76 does; against a control matched on those raw inputs the margin is +11.9 points (CI [+3.3, +20.7] at
77 three seeds). Whether its edge over raw δ comes from the subtraction is not resolved at this budget,
78 so we present it as the recommended instantiation, not a finding about the subtraction itself. On the
79 public corpus, a pre-registered prediction fails: high δ at grasp time accompanies failed grasps, and
80 the traces attribute this to operators over-commanding at contact (Sec. 4.3). A 40-line guard on the
81 same telemetry removes 96% of simulated crush events without retraining (Sec. 4.4).

82 We summarize our contributions below:

- 83 • Characterization of servo tracking error as a contact/load proxy with a validity envelope
84 and a resolution-vs-margin design rule (Sec. 3.2, 4.1).
- 85 • A pre-registered six-design observation study at 280 demonstrations: the channel helps as
86 an input and not as an auxiliary prediction target (Sec. 4.2).
- 87 • A pre-registered 16-dataset corpus study associating high δ at grasp with proxy-defined
88 failure (Sec. 4.3).
- 89 • A runtime jaw guard from bus telemetry alone, with paired deleted-success accounting
90 (Sec. 4.4).

91 2 Related Work

92 **Sensorless force estimation.** Momentum-residual observers recover external torque on torque-
93 controlled arms [12, 13]; virtual force sensors extend the idea to low-cost platforms [14], and
94 Hwang et al. [11] estimate torque from commanded and measured positions on hobby servos, prior
95 art for the raw channel — though their estimator is built on a dedicated per-servo dynamic system
96 identification, so while its inference inputs are positions, the model it needs cannot be obtained
97 from an archived log without access to the hardware that produced it; series-elastic actuation made
98 deflection-as-force a design principle [15, 16]. NeuralActuator [17] learns per-actuator models for
99 external-force perception on this servo class, and FACTR 2 [10, 18] feeds estimated external torque
100 into behavior cloning on commodity arms. Our lag excess is a deliberate special case of FACTR 2’s
101 central idea, a linear free-motion baseline fit on positions alone; the restriction makes the signal
102 retroactive, since every method above consumes live current or load telemetry, a calibration stage,
103 or both. Their own input ablation is the closest independent test of our premise: comparing histories
104 of joint positions, of positions and velocities, and of positions, velocities and tracking error, they re-
105 port that the last “consistently outperforms the other inputs, suggesting that tracking error provides
106 useful information about controller effort and actuator response.” That is measured on hardware, and
107 it concerns the inputs to a force estimator rather than to a policy — but it is the estimator-side form
108 of the question our position-history control asks of a policy, namely whether joint history already
109 subsumes the tracking-error channel. Closest to this work, Wong et al. [8] identify the same channel
110 and test it by subtraction: across four real-world force-critical tasks — surface following, insertion,
111 stiffness discrimination, force-based stopping — they remove the leader–follower discrepancy from

ACT’s action labels and report severe failures in exactly the force-critical phases, restoring competence with torque proxies drawn from motor current or joint effort. That is the real-hardware form of the observability claim our grid can only make in simulation, and it establishes the core of that claim independently of us. Two differences remain, and they are the ones that matter for a fleet of archived data. Their intervention is subtractive, establishing that the cue matters; ours is additive, asking which function of it a small-data policy should observe. The hardware class differs too: they evaluate on an ALOHA-class pair with three cameras including a wrist-mounted view, where we target arms an order of magnitude cheaper whose archived logs carry one scene camera or none. And their remedy estimates joint torque from motor current, $\tau = k_t I$ — telemetry the recording stack caps on the contact joint (App. D), that none of the archived datasets in our corpus records, and whose retroactive recovery from such logs they do not address — while the channel studied here needs two position fields every one of those logs already stores.

Xie and Correll [9] temper this: they judge that outside a few task classes “the performance of imitation learning models is not at a level of dynamics where force truly matters”. The exceptions they name are pouring precise volumes, tight-tolerance insertion, and grasping fragile and deformable objects — the last being the class our force-band task belongs to. We read that as bounding where a force channel pays rather than as disputing that it does, and we adopt the bound: the grid measures observation design on a force-critical task, not manipulation in general.

Observation design in imitation learning. ACT [1], Diffusion Policy [2], OpenVLA [4], π_0 [5], SmolVLA [6], and GR00T N1 [7] all condition on measured joint positions at one or two timesteps; none states a rationale for excluding actuator-error signals. The field’s argument against richer observations is causal confusion: past actions invite copycat shortcuts [19, 20, 21]. Our grid places both effects in one table.

Grasp monitoring and runtime safety. Industrial grippers ship grasp verification from actuator feedback as firmware: Robotiq’s object-detection register fires on commanded-vs-actual discrepancy under a current limit, and the two are separable in their firmware — the current limit stops the fingers, while the detection flag is defined on finger position against requested position [22]. That is the observation-and-enforcement split of this paper, shipping in a commercial gripper. The same manual records its failure mode: detection “may not detect an object even if it is successfully grasped”, for instance a thin object in a fingertip grasp — a binary detector cannot separate a light grasp from none, which is the distinction between detecting contact and measuring it. SCHUNK advertises workpiece-loss detection integrated with gripping-force maintenance [23]; tactile alternatives require added sensing [24]. Academic safety filters for learned policies [25, 26, 27] consume a dynamics model, a state estimate, or exteroception; the guard of Sec. 4.4 applies the industrial telemetry-only approach to wrap learned policies, and we have not found a filter in the academic literature that runs from servo-bus telemetry alone.

3 Method

3.1 Problem formulation

Let $q[t]$ denote measured joint positions in integer encoder counts and $g[t]$ commanded goal positions, both logged at every control step (in LeRobot format, `observation.state` and `action`). The tracking error and lag excess are

$$\delta[t] = g[t-1] - q[t], \quad e[t] = \delta[t] - \hat{K} \cdot r[t], \quad r[t] = g[t-1] - g[t-2], \quad (1)$$

where $r[t]$ is the one-step command rate and $\hat{K}$ is a per-joint coefficient fit by least squares through the origin on the free-motion (jaw-open) frames of the training demonstrations, computed once and frozen before training (full computation, alignment convention, and normalization in App. A). Each servo runs a proportional position loop, so joint torque obeys $\tau \approx K_p \cdot \delta$ and jaw grip force obeys $F \approx \kappa \cdot \delta_{\text{jaw}}$ for a calibration constant κ in newtons per count. Two conditions make δ meaningful: the logged action must be the command actually applied to the servo, and command-to-state timing must be known and stable. Both hold by construction in our simulated plant; on real logs they are

empirical properties of the recording stack (Sec. 4.3). We study which function of the logged pair (q, g) a policy trained on $N = 280$ demonstrations should observe.

3.2 The channel and its envelope

On our simulated plant the jaw calibration is $\kappa = 2.00$ N per count, linear while the jaw is seated and quasi-static, with held-out RMSE 4.3 N over 4–78 N. The same fit carries an intercept of -3.6 N. It is not a measured offset: the fitted forces span 4–78 N, so the smallest δ the fit ever saw is near 3.8 counts and $\delta = 0$ lies outside its domain — the intercept is an extrapolation artifact, which is why the proportional law of Sec. 3.1 carries none. Within the fitted range κ converts *differences* in δ to differences in force; it is used here only in the quantization rule below, which depends on the slope alone, and a reader mapping an absolute δ to newtons with κ and no intercept is off by that constant. These constants describe the modeled plant: the simulator implements the servo loop of Sec. 3.1, so its δ -to-force map is clean by construction; real STS3215 hardware adds stiction, backlash, deadband, saturation, and thermal drift. Bench measurement confirms the saturation and characterizes the baseline registers (App. D); the remainder is still unmeasured. What simulation can characterize is the structure of the proxy: outside the seated quasi-static regime the signal is not a load reading, since free motion contributes following lag (which $e[t]$ removes) and fast transport contributes inertial deflection (App. B, Fig. 5). Encoder quantization yields a design rule: a safe-grip window of force margin M requires an observed quantum on the order of M/κ . We state this as an order and not as a threshold. Treated as a necessary condition it is falsified by our own sweep, which retains most of its success at quanta coarser than M/κ in every margin tested; the rule fires conservatively, ahead of the degradation it predicts, by factors of 1.2–3.1 across three margins (Sec. 4.1). What survives is the mechanism: the failure is a cliff, and its location moves with the margin.

3.3 System, task, and data

All experiments except Sec. 4.3 run on a simulated STS3215-class arm (five joints plus a one-DoF jaw, six servos) with a calibrated contact model. The task is pick-and-place of an object whose retention requires grip force above a lift threshold and whose integrity fails above a crush threshold. A privileged-force oracle passes an admission test on this task and, as a negative control, shows that standard rigid-block pick-and-place cannot distinguish force-aware from force-blind control at all. Demonstrations come from a scripted closed-loop teacher (280 episodes, success-filtered) whose grip regulator acts on the same bus-observable tracking error the arms study, with per-episode grip-depth targets drawn from $U[12, 45]$ counts and injected recovery perturbations; no privileged force sensing enters collection, the teacher’s commands are the applied commands by construction, and the observation arms see only functions of (q, g) . Policies are ACT [1], trained for 12,000 steps at 96 px, identical across arms; at this resolution we cannot rule out a camera deformation cue, but the base arm’s floor performance shows any such cue is insufficient alone. Seed variance was measured before any comparison: $\sigma_{\text{seed}} \approx 10$ points at $n = 100$ evaluation episodes, so 5-seed comparisons resolve gaps of roughly 11 points and we claim nothing finer. Evaluation episodes are common-random-number paired: every arm and seed rolls out the identical episode sequence.

3.4 Observation designs

Six arms share the training recipe and differ only in the proprioceptive observation. **Base (A)**. Measured positions $s[t]$ only: the stock LeRobot observation. **History (B)**. $s[t]$ plus the previous action $a[t-1]$: the minimal change that makes δ recoverable in principle, and the arm the copycat literature [19, 20] warns about. **Delta (C)**. $s[t]$ plus raw $\delta[t]$: the channel with its motion confound intact. **Auxiliary target (D)**. $s[t]$ only, with $\delta[t]$ predicted as an auxiliary target through a small head at loss weight 0.1: the same information as supervision rather than input. **Lag excess (E)**. $s[t]$ plus $e[t]$: the channel with free-motion lag subtracted. **Seat token (F)**. $s[t]$ plus a latched binary seat bit from the guard’s frozen stall detector run causally over $(s[t], a[t-1])$: the channel compressed to one bit.

4 Experiments

We evaluate: (i) over what envelope is δ a valid contact/load proxy; (ii) which observation design exploits it at small data; (iii) does the channel predict grasp outcomes on the public corpus?

4.1 Channel behavior on the simulated plant

The calibration of Sec. 3.2 holds in the seated quasi-static regime. Because the simulator implements the servo loop that generates δ , these numbers validate the proxy against our own plant model; their value is in mapping where the proxy’s structure breaks. A stall detector on net travel and lag excess localizes seat events, and constructed adversarial negatives (free-space reversals, transport shakes) do not fire it within the declared envelope. Coarsening the observed channel eventually collapses force-aware grasping, and rerunning the sweep at a different margin moves the cliff accordingly — which is the signature separating a resolution limit from the mere deletion of input bits. The collapse does not occur at M/κ : at a 100 N limit the channel holds 92% and 75% of its baseline placements at quanta of 4 and 8, both coarser than the predicted threshold, and the observed cliff runs 1.2–3.1 $\times$ beyond the prediction across the three margins. The scale constant here is the pre-registered 2.4 N/count of the sweep rather than the $\kappa = 2.00$ of Sec. 3.2; the discrepancy is reported rather than reconciled, since refitting a constant frozen before the sweep ran would defeat the pre-registration. TYPICAL_PEAK_N, a single median standing in for a contact peak-grip distribution spanning 19–56 N, is the likely source, and the rule should be re-derived from that distribution before it is claimed quantitatively.

4.2 Observation design at 280 demonstrations

Table 1 and Fig. 3 give the grid: 6 arms $\times$ 5 seeds $\times$ 100 paired evaluation episodes, identical otherwise. Arm definitions and directional predictions were frozen before implementation, with E vs. B declared the headline comparison; frozen documents, per-seed outcomes, and all runs are in the released repository. Alongside Welch’s t across seed means we report 95% CIs from a hierarchical bootstrap resampling seeds, then episodes.

One property of this evaluation bounds what the grid can show. All cells are scored with the crush limit disabled, so of the two bounds that make the task force-critical — retention requires grip above a lift threshold, integrity fails above a crush threshold — only the lower one is active. Under that setting a policy succeeds by gripping hard enough and is never penalised for gripping harder: detecting contact suffices, and force *magnitude*, which is what distinguishes the tracking-error channel from a generic temporal observation, cannot pay. The choice isolates observation design from enforcement and we would make it again, but it means the headline comparison does not exercise the property the channel is for.

We attempted to exercise it after the fact and report the attempt as a failed prediction. Re-evaluating the existing checkpoints at 120 N and 60 N crush tiers, we predicted in advance that the channel arms would beat the position-history control, since regulating grip into a band requires magnitude and that control carries almost none. They did not: the lag-excess arm places 0 of 50 at both tiers against 8 of 150 pooled for position history, and at 60 N every arm places zero. Crush counts do not separate the arms either — the single lag-excess seed falls inside the three-seed spread of the control at both tiers.

The screen could not have supported the prediction, for a reason already in our own measurements: force bounds in the demonstrations do not bound the clone. A policy trained on grips at or below 40 N still crushed 15 of 30 episodes at the 60 N tier, and every policy at the larger budget crushes heavily at 120 N. Respecting a ceiling the demonstrations never respected is not available to imitation on this data, whatever the observation carries. Testing whether the channel enables force regulation therefore requires training against a crush tier, which we have not run, and that is the clearest single piece of future work these experiments identify.

Table 1: Placement success (%), mean $\pm$ std over 5 seeds (500 paired episodes per arm), with differences against the channel-free base (A) and the action-history baseline (B). Brackets are hierarchical-bootstrap 95% CIs (resampling seeds, then episodes within seeds); an interval containing zero is unresolved at this budget. Best in bold.

Arm	Observation	Success (%)	vs. A	vs. B
A base	$s[t]$	6.2 ± 5.9		
B history	$s + a[t-1]$	13.8 ± 7.7	$+7.6 [-0.8, +16.0]$	
C delta	$s + \delta$	21.8 ± 9.7	$+15.6 [+5.6, +25.2]$	$+8.0 [-2.8, +18.6]$
D aux-target	$s; \delta$ predicted	6.6 ± 4.0	$+0.4 [-6.2, +6.4]$	
E lag excess	$s + e$	26.6 ± 7.8	$+20.4 [+11.6, +29.0]$	$+12.8 [+2.8, +22.6]$
F seat token	$s + \text{seat bit}$	21.8 ± 9.5	$+15.6 [+6.0, +25.2]$	$+8.0 [-2.6, +18.6]$

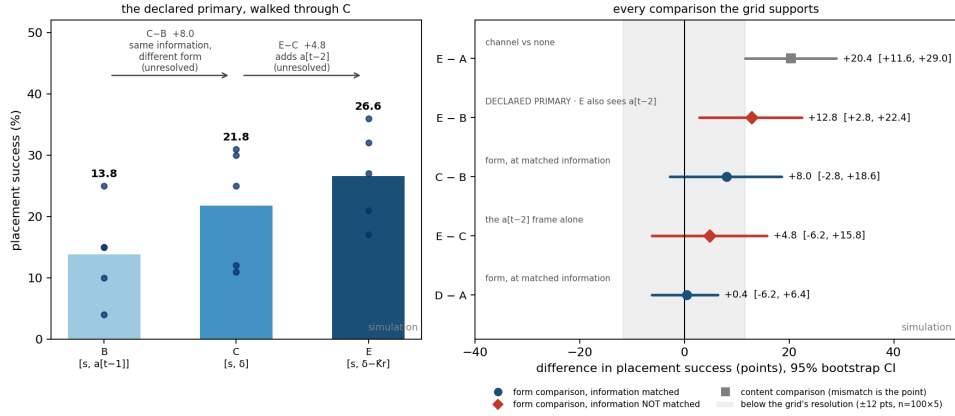


Figure 2: What each comparison in the grid holds fixed. Blue circles vary the *form* of the observation at matched raw content; red diamonds are presented as form comparisons but also differ in content, since the lag excess consumes $a[t-2]$ through the commanded rate and the history baseline does not; the grey square is a content comparison, which is the point of that pair. The shaded band is the below-resolution zone, declared before the grid ran as $15\sqrt{3/n}$ points at $\sigma_{\text{seed}} \approx 10$: 15 points at three seeds, 11.6 at five. Intervals are the hierarchical bootstrap of Table 1, resampling seeds and then episodes within seeds.

256 Three results resolve. First, channel-bearing observation beats channel-free in every input form: C,
257 E, and F each clear the base by at least 15 points with intervals excluding zero (Table 1). Second,
258 the tested auxiliary-prediction design does not recover the direct-input gain: arm D lands equal to base
259 while the same information as an input triples success; other auxiliary weightings and architectures
260 could in principle close this gap, and none was tuned here. Third, in the comparison declared as
261 headline before the confound below was known, only the lag-excess form beats action history at
262 resolution, by 12.8 points (Welch's $t = 2.6$ across seed means, against a bar of 11.6 at five seeds);
263 raw δ 's 8-point margin over history does not resolve. That pair is confounded: E consumes $a[t-2]$
264 through the commanded rate and the baseline does not, so it varies content as well as form, and it
265 is superseded by the information-matched control $B2 = [s[t], a[t-1], a[t-2]]$, which E exceeds by
266 $+11.9$ (three seeds, CI $[+3.3, +20.7]$, below the three-seed bar of 15). We report $E-B$ because it
267 was declared, and rest the form claim on $E-B2$. The remaining pairs stay open, including E against
268 C (difference $+4.8$, CI $[-6.2, +15.8]$), so whether E 's edge comes from the subtraction or from δ
269 at all is not attributable at this budget; resolving $E-C$ by seed expansion would require roughly
270 40–60 seeds per arm at this σ_{seed} , and evaluation is already episode-paired. The frozen prediction
271 named E the sharpest arm and expected D, E, and F to clear B; D's share of that prediction failed,
272 as reported above. An interim 3-seed analysis flagged E 's margin as resolved and F's as at-the-floor,
273 and the expansion to 5 seeds confirmed both calls, collapsing F's margin over B and preserving
274 E 's. Caveats: five seed means per arm, a headline difference of similar magnitude to the declared

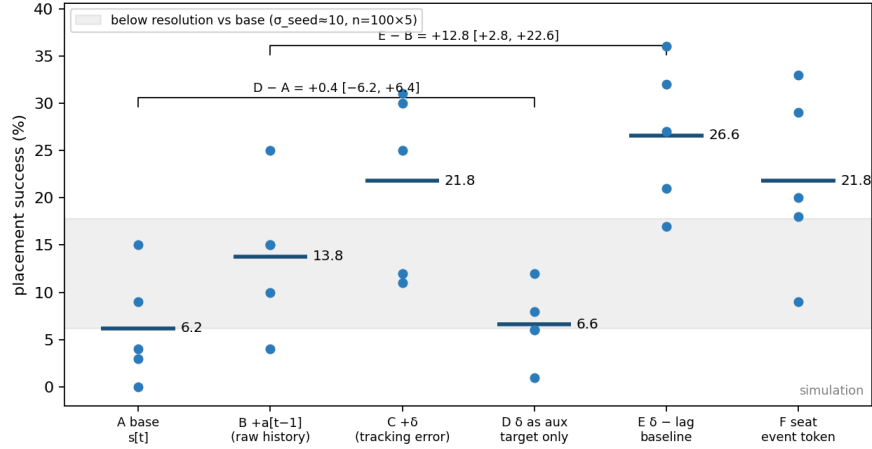


Figure 3: The observation-design grid. Bars are 5-seed means, dots individual seeds (100 episodes each). The gray band spans the below-resolution zone around base given $\sigma_{\text{seed}} \approx 10$. The lag-excess arm (E) is the only design whose margin over the history baseline (B) exceeds seed resolution.

275 resolution, and uncorrected secondary comparisons beyond the designated primary one. A best-
 276 seed robustness probe under object-property shifts (block mass $\times 0.5$ and $\times 1.5$, friction $\times 0.5$; $n=50$
 277 paired episodes per condition, orderings only) shows the channel arms’ margins persist under every
 278 shift while the channel-free base returns byte-identical outcomes in all conditions – the signature of
 279 a policy that never interacts with the object’s dynamics – supporting the reading that channel-bearing
 280 policies regulate contact rather than replay a visual sequence (protocol and per-condition table in the
 281 repository).

282 Within those limits the ordering supports one design statement: at small data on this force-critical
 283 task, observing the channel is the change that moves performance, and the free-motion-compensated
 284 form is the best tested way to observe it.

285 4.3 Ecosystem study: a pre-registered negative

286 Does δ predict grasp outcomes corpus-wide? The protocol was frozen before the run: seat detection
 287 by the stall detector of Sec. 4.1, δ -at-seat as unsigned tracking error, success proxy defined as the
 288 gripper staying closed through transport, and the prediction that successes carry higher δ with Co-
 289 hen’s $d > 0.5$ in at least 10 of 20 datasets. Of the 20 targeted SO-100/SO-101 datasets, 16 passed
 290 the frozen inclusion gates with connectivity: 546 episodes, 283 proxy successes.

291 **The prediction fails: 2/16 datasets** (pooled within-dataset $d = -0.39$). The channel is informative
 292 with the opposite sign: $|d| > 0.5$ in 10/16 datasets, 7/16 significantly inverted (Fig. 4). The traces
 293 explain the physics. Teleop leader devices have no hard stop, so operators over-command every
 294 close; the seat flag fires in nearly every episode of 14/16 datasets because empty jaws stall at the
 295 mechanical limit exactly as loaded ones do. Successful grasps are stereotyped low- δ closes (in one
 296 dataset every clean episode rests at the same 7-count gap), while failures carry 40–80-count frames
 297 of an operator forcing a jammed grasp. High δ at grasp time is associated with failure under the
 298 frozen proxy. This rules out mining the corpus for success labels through this channel alone, and it
 299 supports enforcement: the high- δ events the corpus associates with failure are what the guard caps.

300 The result also bounds Sec. 4.2’s transfer to retroactive training on real teleop data. When the seat
 301 event fires in essentially every episode, a latched seat bit (arm F) carries no information, and raw
 302 δ at close time reads commanded depth as much as contact; what stays informative is δ magnitude
 303 and its dynamics, which is what the inverted effect sizes measure. Whether the lag-excess form
 304 retains its simulated advantage when trained on over-commanded human demonstrations is an open
 305 empirical question, not a corollary. Two caveats attach to the corpus numbers: the recorded action

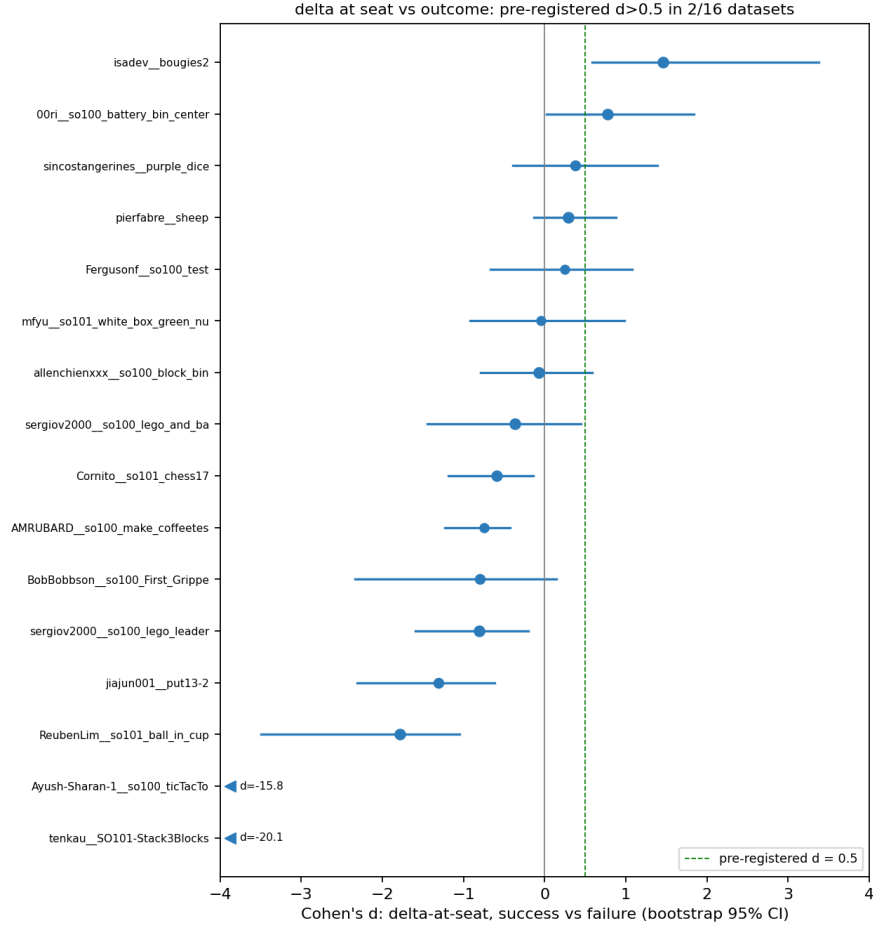


Figure 4: Per-dataset Cohen’s d of δ -at-seat between proxy successes and failures, bootstrap 95% CIs, for the 16 datasets passing the frozen gates (attrition documented in the repository). Positive d was the pre-registered direction. Two datasets exceed $d = 0.5$ as predicted; six are significantly inverted; two task-repetitive datasets reach $d \approx -16$ and -20 because their stereotyped successes have near-zero δ variance.

306 is the leader position, which approximates but need not equal the applied follower command, and
 307 action-to-state timing is characterized only up to an integer-offset sweep over $k \in \{0, \dots, 5\}$, which
 308 leaves these conclusions unchanged (the median per-dataset shift in Cohen’s d across offsets is 0.12)
 309 while estimating the physical alignment at three to four recorded frames (App. A).

310 4.4 Runtime guard

311 The same telemetry supports enforcement. A 40-line jaw guard (net-travel stall and lag-excess seat
 312 detection, a 12-count squeeze budget past seat, a closing-rate limit) runs from bus signals with no
 313 dynamics model, no retraining, and no object knowledge; all results are simulated. Paired its detec-
 314 tion thresholds make it insensitive to closes slower than its rate gates, so it engages on aggressive
 315 closers and not on the gentle scripted teacher (App. C), and on three learned checkpoints over iden-
 316 tical paired episodes its first-generation detector (App. C) reduced crush events from 85 to 3 of 180
 317 (96%) across the crush-capable tiers, with median peak force over those same episodes dropping
 318 from 33–92 N to 13–26 N. It also removes every success in the experiment: 12 of 270 unguarded
 319 become 0 of 270, and no guarded cell scores above zero. The policies are weak to begin with here
 320 (4.4% unguarded, and the channel-free arm scores zero either way), so this is the guard deleting

the few successes that existed rather than dismantling a working policy — but a wrapper that never succeeds is not yet a deployable one. The shape of the trade is not peculiar to us: Nakamura et al. [28] report a least-restrictive latent safety filter on hardware that drives failures from 62% to zero while stalls rise from zero to 62% and task success is unchanged at 38%, converting every failure into a stall and adding nothing. Ours is the more severe instance of that mode, since it removes the successes as well, and the comparison names what a deployable version needs rather than excusing what ours does: intervention selective enough to leave the succeeding trajectories intact. Demonstrations must record applied commands, and wrapped controllers must be closed-loop on contact. Details, figure, and integration accounting in App. C. On hardware the register this guard would otherwise read is capped by the recording stack itself and saturates through contact, while the channel does not (App. D).

5 Limitations

All learning and guard results are simulation, on one gripper and one object family; every quantity in newtons inherits the simulator’s servo and contact model: the hardware appendix settles which bus signal carries contact, and does not put a newton on it (App. D). The task is constructed to require force-band control, so the grid measures observation design on a force-critical task, not manipulation in general. The lag-excess model is a first-order correction; an offline ladder of richer free-motion predictors (intercept, direction and acceleration terms, short causal FIR) improves held-out free-motion residual by at most 1.9% over the scalar model on the jaw, so the linear choice is not leaving obvious compensation on the table, though learned predictors in the style of Oh et al. [10] remain untested as policy inputs. The seed budget leaves $C-B$ and $E-C$ open; the subtraction is addressed below against an information-matched control rather than by those pairs.

The most consequential open question concerns operating point. The grid runs at 280 demonstrations, 96 px and 12k steps. At a larger budget — 600 demonstrations, 224 px, 50k steps, where absolute success reaches 57–63% rather than the 6–27% reported here — a two-seed comparison of the raw channel against the action-history baseline is indistinguishable (60% vs. 57%, with a within-policy seed spread of 12 points, four times the between-policy gap). That cell tested C against B ; E , the design this paper recommends, has not been run at that budget. Two readings are therefore live and we cannot currently separate them: that the channel’s advantage is a small-data effect which disappears where policies become useful, or that the free-motion subtraction is precisely what survives scaling and the raw form is not. What the small-data ladder does support independently of either is that policies with no action-history input in any form plateau at 16–24% lift commitment across three data recipes while channel-bearing policies reach 47–76%: the information appears necessary, and it is the *form* whose value at scale is unresolved. This pattern is not peculiar to our setup: Zeng et al. [29] sweep context length against dataset size and find the ordering between short- and long-context policies reverses between 200 and 1000 demonstrations, with dataset size disproportionately favouring the longer-context policy. Their own recommendation applies directly to us: they observe that datasets of 50–100 demonstrations remain common and “encourage evaluating results across dataset sizes to ensure results hold at scale”. Our 280-to-600 window sits inside their crossover, so the tie we observe is the expected behaviour of a known effect rather than an unexplained inconsistency — which also means a single budget cannot identify the mechanism, and any arm we compare must be compared at more than one.

The next caveat bears on the comparison declared as primary. The lag excess $e[t] = \delta[t] - \hat{K}r[t]$ depends on $a[t-2]$ through the commanded rate, while the action-history baseline B observes $s[t]$ and $a[t-1]$ only, and $\hat{K}$ is far from zero (0.74–0.84 across joints, 0.80 on the jaw). E therefore consumes one more frame of action history than the baseline it is compared against, and $E-B$ confounds the free-motion subtraction with that additional frame. The comparison that isolates the subtraction holds the raw inputs fixed: E against a two-step raw history $B2 = [s[t], a[t-1], a[t-2]]$, information-matched to E by construction. We ran it at three seeds under a rule frozen beforehand. $B2$ reaches 14.7 ± 3.2 and E exceeds it by $+11.9$ ($t = 3.03$, CI $[+3.3, +20.7]$): against a control with identical raw inputs the advantage persists, which answers the confound directly. We

decline the available decomposition. $B2$ over the one-frame baseline is $+0.9$ with CI $[-7.7, +9.0]$, a null wide enough to be consistent with the extra frame carrying most of $E-B$ or none of it, so the point estimates do not partition $E-B$ and we do not report them as though they did. Neither difference clears the 15-point bar declared in advance, and $B2$ is 18-dimensional against E 's 12, so the comparison is matched in information and not in width.

A wider audit of the arms' information dependence sharpens this: of the design comparisons the grid draws conclusions from, only $C-B$ and $D-A$ hold raw content fixed while varying form. F in particular consumes an entire episode prefix, because its seat detector latches once and never resets, yet does not beat E — so a simple information ordering does not explain the grid either. $D-A$ is matched by construction, the channel entering only as an auxiliary target, which makes the observe-versus-predict null the grid's most securely grounded design result. Nothing here bears on the channel-versus-channel-free result ($E-A = +20.4$, $t = 4.7$); what is unresolved is the narrower claim that the lag excess specifically improves on action history. Running E and the matched two-step baseline at both budgets is the experiment we would prioritize.

A further control bears on the mechanism, and it does not favour the strong reading. Arm G observes $[s[t], s[t-8]]$ — joint history only, from which δ is not computable even in principle — and is width-matched to E . At three seeds it reaches 20.3 ± 4.7 . It is indistinguishable from the raw-channel arm C (-1.5 , $t = -0.29$) and from E itself ($E-G = +6.3$, CI $[-3.2, +16.0]$), while clearing the channel-free base by $+14.1$ ($t = 3.73$, CI $[+5.9, +22.1]$) — 69% of E 's own margin over that base. The ordering across arms is therefore base 6.2, position history 20.3, raw channel 21.8, lag excess 26.6: the large step is from no temporal information to some, and the step from generic history to this channel does not resolve at this budget. Under the declared bar, $E-G$ would need roughly 18 seeds.

We state the consequence plainly. What this grid establishes is that an observation beyond $s[t]$ is necessary; *which* form it must take is open, and the reading on which the servo tracking error specifically is the enabling observation is not established by these experiments. That proposition and the long-context account of Zeng et al. [29] are both consistent with what we measured. We do not claim that position history fails to substitute for the channel; at three seeds we cannot distinguish the two, which is a different statement and the one the success rates support. The observations themselves are not equivalent. Measured against ground-truth grip force over the 80,907 demonstration frames where an eight-step lookback exists, the joint-history signal detects contact markedly worse than the tracking error (AUC 0.78 against 0.95, the direction inverted for joint history since a blocked jaw moves *less*) and carries far less force magnitude ($R^2 = 0.12$ against 0.76 within contact frames). Whatever produces G 's success, it is unlikely to be a force estimate. $C-B$ is by contrast a clean representation test, since C and B observe the same raw quantities — and it does not resolve at our seed budget ($+8.0$, $t = 1.44$). The decomposition is exact: $E-B = (E-C) + (C-B) = +4.8$ ($t=0.86$) $+ 8.0$ ($t=1.44$) $= +12.8$, so the one pair clearing our resolution bar is significant only as the sum of two components neither of which is separately claimable at $n = 5$ seeds. On real logs, δ depends on action semantics and timing that the corpus study characterizes only up to an integer-offset sweep. The corpus success proxy is unvalidated against ground truth, and multi-grasp tasks were excluded by design. The grid is measured on one demonstration collection; a three-seed recollection (App. C) surfaces collection-level resampling variance that our comparisons do not yet average over. The channel is valid only seated and quasi-static.

6 Conclusion

Position-command logs contain a retrospectively computable contact proxy: the servo tracking error, valid within a quasi-static envelope our simulated plant makes measurable. At 280 demonstrations, policies must observe this channel; predicting it as the tested auxiliary target recovers none of the gain, and the free-motion-compensated form is the best tested instantiation. Across 16 public datasets high δ at grasp time accompanies failure, which rules out free outcome labels from this

channel alone and supports its use for enforcement. What remains open requires hardware: bench calibration, retroactive training on real teleop logs, and a copycat-style probe of when history helps versus hurts in behavior cloning.

References

- [1] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Robotics: Science and Systems (RSS)*, 2023.
- [2] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 2023.
- [3] R. Cadene, S. Alibert, A. Soare, Q. Gallouedec, A. Zouitine, and T. Wolf. Lerobot: State-of-the-art ai for real-world robotics. <https://github.com/huggingface/lerobot>, 2024. Source audited May 2026.
- [4] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn. Openvla: An open-source vision-language-action model. In *Conference on Robot Learning (CoRL)*, 2024.
- [5] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- [6] M. Shukor, D. Aubakirova, F. Capuano, P. Kooijmans, S. Palma, A. Zouitine, M. Aractingi, C. Pascal, M. Russi, A. Marafioti, et al. Smolvla: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025.
- [7] NVIDIA, J. Bjorck, F. Castaneda, N. Cherniadev, X. Da, R. Ding, L. Fan, Y. Fang, D. Fox, F. Hu, et al. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025.
- [8] K. H. Wong, L. Liu, and F. Dayoub. Beyond implicit force: Evaluating explicit force-torque proxies in action chunking with transformers. *arXiv preprint arXiv:2607.14578*, 2026.
- [9] W. Xie and N. Correll. Towards forceful robotic foundation models: a literature survey. *arXiv preprint arXiv:2504.11827*, 2025.
- [10] S. Oh, J. J. Liu, T. Tao, P. Han, K. Shaw, S. Funabashi, R. Salakhutdinov, and D. Pathak. FACTR 2: Learning external force sensing for commodity robot arms improves policy learning. *arXiv preprint arXiv:2606.12406*, 2026.
- [11] Y. Hwang, Y. Minami, and M. Ishikawa. Virtual torque sensor for low-cost RC servo motors based on dynamic system identification utilizing parametric constraints. *Sensors*, 18(11):3856, 2018.
- [12] A. De Luca and R. Mattone. Sensorless robot collision detection and hybrid force/motion control. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2005.
- [13] S. Haddadin, A. De Luca, and A. Albu-Schäffer. Robot collisions: A survey on detection, isolation, and identification. *IEEE Transactions on Robotics*, 33(6):1292–1312, 2017.
- [14] S.-H. Yen, P.-C. Tang, Y.-C. Lin, and C.-Y. Lin. Development of a virtual force sensor for a low-cost collaborative robot and applications to safety control. *Sensors*, 19(11):2603, 2019.
- [15] G. A. Pratt and M. M. Williamson. Series elastic actuators. In *Proceedings 1995 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 399–406, 1995.

- [16] M. M. Williamson. Series elastic actuators. Master’s thesis, Massachusetts Institute of Technology, 1995.
- [17] Z. Dou, J. U. Onyemelukwe, H. Zhang, H. Zhang, M. Guo, Y. Tian, M. P. Lipiec, J. Jacob, C. Liu, P. Y. Chen, Y. Ivanov, and W. Matusik. Neuralactuator: Neural actuation modeling for robot dynamics and external force perception. In *Robotics: Science and Systems (RSS)*, 2026. arXiv:2607.11734.
- [18] J. J. Liu, Y. Li, K. Shaw, T. Tao, R. Salakhutdinov, and D. Pathak. FACTR: Force-attending curriculum training for contact-rich policy learning. *arXiv preprint arXiv:2502.17432*, 2025.
- [19] P. de Haan, D. Jayaraman, and S. Levine. Causal confusion in imitation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [20] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao. Fighting copycat agents in behavioral cloning from observation histories. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [21] M. Torne Villasevil, A. Tang, Y. Liu, and C. Finn. Learning long-context diffusion policies via past-token prediction. In *Conference on Robot Learning (CoRL)*, 2025.
- [22] *2F-85 and 2F-140 Instruction Manual*. Robotiq, 2019. Control section: gOBJ object-detection register.
- [23] *EGK Electric Parallel Gripper Product Documentation*. SCHUNK, 2024. Workpiece-loss detection.
- [24] R. Calandra, A. Owens, M. Upadhyaya, W. Yuan, J. Lin, E. H. Adelson, and S. Levine. The feeling of success: Does touch sensing help predict grasp outcomes? In *Conference on Robot Learning (CoRL)*, 2017.
- [25] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu. Safe reinforcement learning via shielding. In *AAAI Conference on Artificial Intelligence*, 2018.
- [26] K.-C. Hsu, H. Hu, and J. F. Fisac. The safety filter: A unified view of safety-critical control in autonomous systems. *Annual Review of Control, Robotics, and Autonomous Systems*, 2024.
- [27] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig. Safe learning in robotics: From learning-based control to safe reinforcement learning. *Annual Review of Control, Robotics, and Autonomous Systems*, 5:411–444, 2022.
- [28] K. Nakamura, A. L. Bishop, S. Man, A. M. Johnson, Z. Manchester, and A. Bajcsy. How to train your latent control barrier function: Smooth safety filtering under hard-to-model constraints. *arXiv preprint arXiv:2511.18606*, 2025.
- [29] M. Zeng, A. Agarwal, A. Bati, B. Lee, S. Ancha, and R. Tedrake. Revisiting open-loop execution in robotics: Toward reactive, higher-performing policies. *arXiv preprint arXiv:2608.15938*, 2026.

500 A Channel computation and implementation details

501 All signals are in integer encoder counts at the recorded cadence (stride-2 frames of a 30 Hz bus,
 502 i.e. 15 Hz): $\delta_j[t] = a_j[t-1] - s_j[t]$, where a is the commanded and s the measured position, the
 503 $k=1$ alignment convention. An offset sweep over $k \in \{0, \dots, 5\}$ on the public corpus leaves
 504 every Sec. 4.3 conclusion unchanged (the median per-dataset shift in Cohen’s d across offsets is
 505 0.12) while estimating the physical alignment at three to four recorded frames; the bench latency
 506 measurement (App. D) will pin it. The commanded rate is the first difference of commands, $r_j[t] =$
 507 $a_j[t-1] - a_j[t-2]$, zero at the first two frames of an episode. The lag model is a single per-joint
 508 scalar through the origin, $\hat{K}_j = \sum r\delta / \sum r^2$, fit over frames whose jaw command exceeds its 30th
 509 percentile; the jaw is the task’s only contact, so jaw-open frames are free motion. Fitted values
 510 span $\hat{K} = 0.74\text{--}0.84$ across joints on our training set (and move by at most 0.09 when the free-
 511 motion percentile criterion is varied from the 10th to the 50th), computed once on the training
 512 demonstrations and frozen. The policy input is the concatenation $[s_{\text{norm}}, (\delta - \hat{K}r - \mu_\delta)/\sigma_\delta]$; joint
 513 states and actions are standardized by dataset statistics, while δ , e , and the aux target use a fixed scale
 514 of $\sigma_\delta = 8$ counts with $\mu_\delta = 0$, so no channel statistic leaks from evaluation. The same statistics
 515 are saved beside the checkpoint and applied identically at evaluation, where $a[t-1]$ is the previously
 516 issued command, so the feature is computable from exactly what the deployed stack already has.
 517 Appended channels concatenate onto the state vector, leaving the network otherwise unchanged.

518 B Mechanism on a single grasp

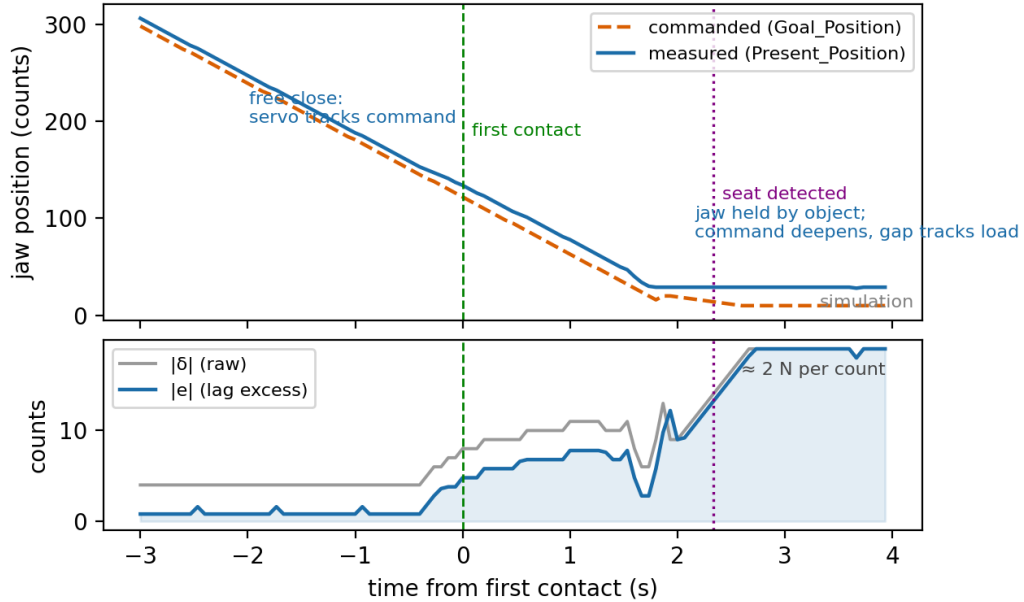


Figure 5: The channel on one servo and one grasp. The commanded position (dashed) and measured position (solid) track closely through the free close, separated only by the small following lag. At first contact (marked) the measured position stalls while the command continues to deepen; the growing gap is the load reading, $\kappa \approx 2$ N per count on the simulated plant. The detector’s seat event (marked) fires when net travel stalls while lag excess grows. The lower panel plots $|\delta|$ and the lag excess $|e|$ over the same frames; $|e|$ sits near zero through the free close by construction and inherits the load reading after contact.

C Guard details

The guard combines net-travel stall and lag-excess seat detection, a squeeze budget of 12 counts ($\approx +24$ N on the simulated plant) past seat, and a closing-rate limit. Two regimes follow from the detector’s thresholds. On aggressive closers it binds: learned policies commanding fast or deep closes are capped near 40 N against raw peaks of 99–127 N, and an open-loop clamp reference commanding 360 N grips is held to 10–28 N. On the gentle closed-loop teacher it never fires, and all three of its mechanisms have to be accounted for: the teacher’s ~ 0.3 counts per frame in its fine pass sits below both detection gates, and the closing-rate limiter — which applies unconditionally, before and independent of seat detection — is set at 4 counts per policy step against roughly 0.6 demanded, a factor of seven clear. A paired run shows 25/30 placements wrapped versus 28/30 raw at statistically indistinguishable ~ 39 N grips: no mechanism bound, so the run measures wiring overhead and not a cost of enforcement. Whether threshold-level insensitivity to gentle closes is acceptable conservatism or a coverage gap depends on the deployment’s hazard model; we state it as a measured property of these thresholds.

An independently recollected demonstration set (same teacher, recipe, and budget; fresh episodes) trained at three seeds per arm neither resolves any per-arm difference from the original grid nor reproduces its point-estimate ordering at that reduced seed count (the lag-excess arm’s fresh-collection seeds span 7–27 against the original 17–36, Welch’s $t = 1.6$). Across the six arms the three channel-bearing designs’ point estimates fell while the three channel-free designs’ rose, none individually resolving; four marginal statistics of the two collections (tracking-error distributions, command floors, command-rate distributions, per-episode grip-depth diversity) are indistinguishable, so we treat this as unattributed collection-level resampling variance – a component above seed variance that the headline comparison of Sec. 4.2 does not yet average over, and that hardening it requires spanning with seed expansion over multiple collections. The crush-elimination A/B predates the detector’s final revision: on three learned checkpoints over identical paired episodes the first-generation detector (single-step stall test, 4-count budget) reduced crush events from 85 to 3 of 180 across the two crush-capable tiers (Fig. 6), with median peak force over those same 180 episodes dropping from 33–92 N to 13–26 N; across all nine cells, including the no-crush tier where crushing is impossible, the unguarded range is 33–131 N. A rerun under the final detector is pending alongside hardware validation. The guard also deletes successes, and it deletes all of them: unguarded totals are 12 successes in 270 episodes, guarded totals are 0 in 270, and every one of the nine cells scores 0/30 guarded. The losses fall on two arms — the quantized arm gives up 7/30 at a 131 N median and 1/30 at the 120 N tier, the delta arm 2/30 and 2/30 — while the channel-free arm scores zero with and without the guard. Since the unguarded rate is 4.4%, this is a guard removing the few successes that existed rather than a working policy being dismantled, but it means the enforcement layer as tested trades all measured task success for a 96% crush reduction. Integration requires two properties: demonstrations must record applied commands rather than intentions, and any wrapped controller must be closed-loop on contact, since the rate limiter breaks open-loop grasp timing by construction. Collecting demonstrations under *active* guard enforcement would require a detector variant sensitive to gentle closes and remains untested.

D Hardware: the reviewer’s baseline on a physical STS3215

The first question this paper invites is why estimate contact from tracking error when the servo already reports `Present_Load` and `Present_Current` over the same bus. This appendix answers that question on hardware and states plainly what it does not answer: the newton scale is not yet measured, so every force quantity in the body still inherits the simulator’s contact model.

D.1 Static: the register is the channel, rescaled

Protocol. A payload staircase on one STS3215 in an SO-101 shoulder-lift joint. Twelve hanging masses spanning 63.7–319 g, each held at five arm poses, logging `Present_Load` (address 60),

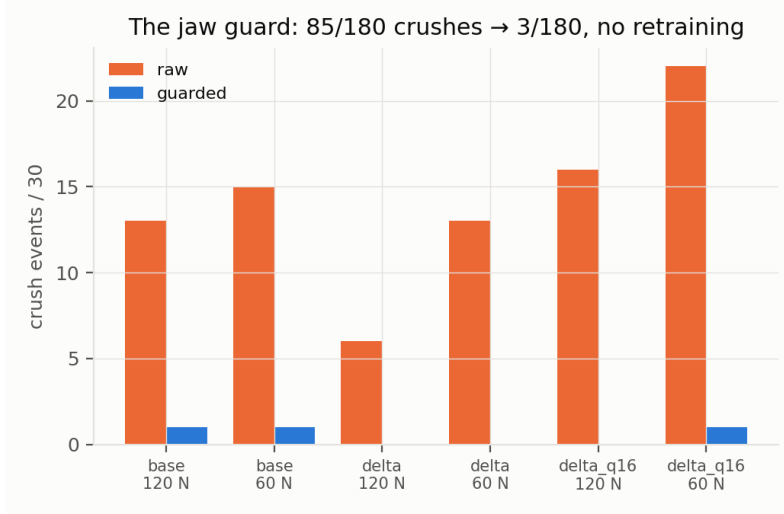


Figure 6: Crush events per 30 paired episodes for three learned checkpoints (base, delta, and a 16-count-quantized delta arm) at two crush tiers, raw (orange) versus guarded (blue), under the first-generation detector. Identical checkpoints and episode seeds; the guard acts only through bus telemetry at runtime. Totals: 85 of 180 raw, 3 of 180 guarded.

567 Present_Current (69), Present_Position (56) and Goal_Position (42) at static equilibrium;
 568 $\delta = \text{Goal} - \text{Present}$, in encoder counts. Masses are known from package labelling rather than
 569 weighed, so their systematic error is unquantified. Three masses were measured twice (106, 149,
 570 191 g), and those replicates are used below as the repeatability estimate; they bound the random
 571 component only.

572 **Poses are fit separately.** Pooling them regresses across five different lever arms at once and inflates
 573 every residual. All intervals below are 95% bootstrap percentiles over 10,000 resamples of the
 574 twelve masses.

pose	δ slope (counts/g)	R_δ^2	R_{load}^2	R_{curr}^2	replicate $ \Delta_\delta $
3	-0.0194 ± 0.0019	0.910 [0.85, 0.98]	0.910	0.014	0.33
1	-0.0323 ± 0.0044	0.845 [0.72, 0.95]	0.845	0.726	1.33
2	-0.0435 ± 0.0145	0.473 [0.10, 0.81]	0.473	0.503	7.00
0	-0.0236 ± 0.0126	0.261 [0.01, 0.71]	0.279	0.041	4.00
4	-0.0115 ± 0.0119	0.085 [0.00, 0.63]	0.077	0.034	2.67

Table 2: Per-pose fits against true hanging mass, ordered by R_δ^2 . Replicate $|\Delta_\delta|$ is the mean absolute difference in δ between the two independent measurements at each repeated mass, in counts.

575 **The register carries the same information as the channel.** Pooled over all sixty measurements,
 576 $\text{Present_Load} = -4.70 (\pm 0.10) \delta + 0.1$ with $R^2 = 0.976 [0.969, 0.982]$ (Fig. ??, right). Per pose
 577 the two track the true mass to within 0.017 of each other in R^2 , agreeing to three decimals in three
 578 of five poses. They are one signal on this joint, so the register cannot be a better instrument than the
 579 channel — only a differently scaled one. This is the claim the appendix is for, and it is the one the
 580 data supports most strongly.

581 **What limits it is repeatability, not lever arm.** The spread in Table 2 is large, and it does not follow
 582 slope magnitude: pose 2 has the steepest response to mass and still resolves poorly, while pose 3 has
 583 a middling slope and the tightest fit. It follows the replicate column instead. Dividing repeatability
 584 by slope gives the mass a single measurement can distinguish: about 17 g at pose 3 and 41 g at pose
 585 1, against 160–230 g at the three poorly repeating poses. The channel’s usable resolution on this
 586 joint is therefore a property of where the arm is standing, not a single number, and the favourable
 587 figure is the exception rather than the summary.

588 **Current is not established as the better instrument, or the worse one.** δ attains the higher R^2
589 in four of five poses, which under an exact paired sign test is $p = 0.375$ — at $n = 5$ poses that is
590 uninformative in either direction, four coin flips landing the same way. We report the ordering and
591 claim nothing from it.

592 **D.2 Dynamic: the baseline loses its range exactly at contact**

593 Fifty teleoperated demonstrations on the physical arm pair (21,476 frames, a pick-and-place with a
594 single overhead camera and no wrist view) show what the static staircase cannot. `Present_Load`
595 pins at 500 in 16.5% of frames, and all fifty episodes reach it. The saturation is not spread evenly:
596 it coincides with contact, where $|\delta|$ averages 12.35° against 2.09° elsewhere. Inside those frames
597 the register reports a single constant value while δ still spans 8.01 – 31.01° , twenty-three degrees of
598 range the baseline cannot express. Where the register is not clipped the two agree as the static data
599 predicts ($r = -0.853$ unsaturated, -0.922 overall).

600 **D.3 The ceiling belongs to the stack, not to this arm**

601 Reading the torque-limit register (address 48) returns 1000 for joints one through five and 500 for
602 the gripper alone. `Present_Load` is full-scale 1000, so 500 is a configured cap on the one joint
603 that makes contact. It is not a local misconfiguration: LeRobot [3] writes it for every SO-100/SO-
604 101 follower it connects, inside the configuration path that connection always invokes, together
605 with a matching halving of the current-protection register — both justified, reasonably, as burnout
606 protection.

607 Both baselines are therefore capped, on the contact joint, by the library the fleet records through. δ
608 requires no register and no limit; it is arithmetic over two numbers every one of those logs already
609 stores, and is untouched by the same configuration.

610 The register read is from one arm, and we state the limit rather than leave it inferred: a single
611 follower reading 500 on its gripper is equally consistent with that unit having been flashed that way.
612 The load-bearing evidence is therefore the code path, which writes the value unconditionally on
613 every connection, not the measurement, which confirms the value is present on the one unit we have.
614 We ran the discriminating test and it did not discriminate. The leader configuration path performs no
615 torque-limit write, so a leader gripper at full scale beside a follower gripper at 500 would have shown
616 the stack writing the cap; instead both grippers on our pair read 500, with joints one through five
617 reading 1000 on each. Two explanations survive and our hardware cannot separate them: the units
618 may ship with the gripper capped, or the leader may at some point have been connected through a
619 follower class — a port swap this bench’s own tooling explicitly anticipates — which would have
620 written the value and erased the evidence permanently.

621 The claim therefore rests on the source rather than on our measurement. LeRobot writes 500 on
622 every follower it connects, which any reader can check in the configuration path, and our registers
623 are consistent with that without confirming it. A clean test needs a follower that has never been
624 connected through the library, and we did not have one.

625 **D.4 What is still not measured**

626 Four gaps, stated because they bound the claims above.

627 No load cell has been applied, so the newton scale is unmeasured and the simulator’s 2.0N/count
628 remains the provenance of every force figure in the body. Hanging masses give torque in the gravity
629 direction on one joint; they do not give newtons per count in the jaw’s closing direction, which is the
630 quantity the task actually depends on. The static staircase is one joint, leaving stiffness measured
631 at $n = 1$ and its per-servo-versus-per-joint prediction untested. It is static only: no three-speed
632 sweep, no opening and closing branches for hysteresis, no drift checks, and no command-to-state
633 latency distribution — the last of which Sec. 3.1’s alignment argument still resolves by corpus
634 sweep rather than by measurement. Finally, the current half of the saturation result is inferred from

635 the configuration write rather than observed, since these recordings log position and load but not
636 current.

637 Comparison rows for a completed calibration remain the simulator's 2.0 N/count, Hwang et al. [11],
638 and NeuralActuator [17].